

Mechanism-checked error allowance

New audit | 21 September 2026

The previous 100% result survives direct checks of the proposed growth mechanism at the measured powers. Ablations show that correcting the boundary condition accounts for most of the coverage gain. They do not identify the KZ exponent as uniquely responsible.

Ablation: what actually supplies coverage?

Allowance	Pooled coverage	Trial success / CP lower	Mean S/e
Boundary only	0.164	0/3 / 0.000	0.390
Fitted slope + boundary	0.997	2/3 / 0.135	0.209
KZ slope only	0.296	0/3 / 0.000	0.402
KZ slope + boundary	1.000	3/3 / 0.368	0.240
Reference 0 + boundary	1.000	3/3 / 0.368	0.218
Reference 1 + boundary	1.000	3/3 / 0.368	0.260

348 configurations reuse three distinct test trials. Trial success means all configurations for that trial passed. CP values are one-sided 95% nominal binomial lower bounds, not intervals on the pooled rate. Same MLP(8), seed 0 and legacy LOTO b as the prior ablation; no models were retrained.

Does the proposed growth allowance hold?

Law configuration	Endpoint checks	Interval checks	Largest interval ratio
Ceiling 146 mW	4/4	4/4	0.741
Ceiling 192 mW	3/3	3/3	0.696
22-split sweep	29/29	29/29	0.696

Growth is independent of anchor, so 12 repeated anchor checks were removed. The 29 sweep checks reuse three test trials and several intervals; they are not 29 independent observations. Ratios below one support the assumed envelope on measured secants only.

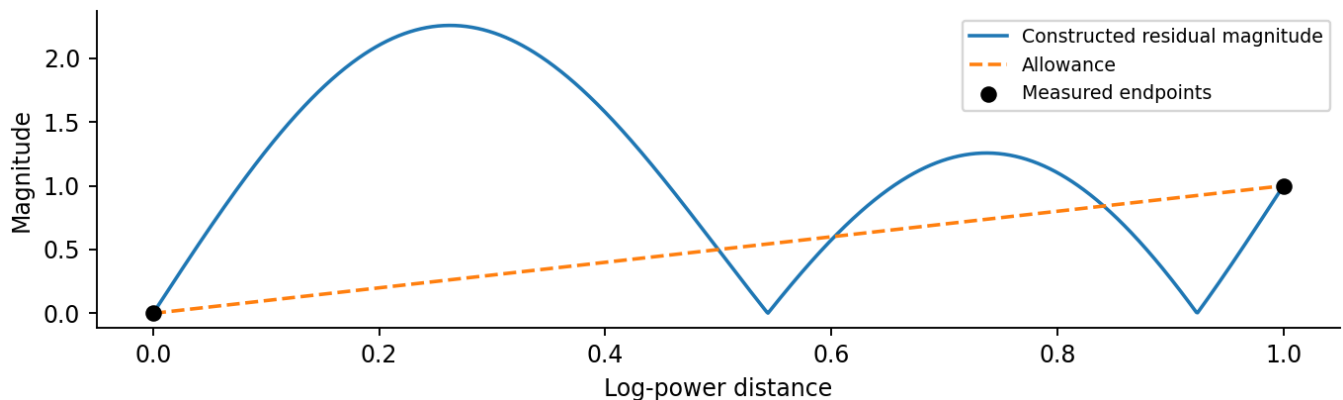
What this supports, and what it does not

The exact decomposition

Let r_0 be the residual at the calibration ceiling, δ the log-power distance, q the fitted law slope and q_{sec} the measured ceiling-to-test slope. The identity $r = r_0 + \delta(q - q_{\text{sec}})$ holds numerically for every audited configuration. The proposed allowance is $\text{norm}(r_0) + k\delta$.

Its margin equals growth slack plus triangle-inequality slack: $[k\delta - \text{norm}(\delta(q - q_{\text{sec}}))] + [\text{norm}(r_0) + \text{norm}(\delta(q - q_{\text{sec}})) - \text{norm}(r)]$. The first term is nonnegative in every measured check, so endpoint coverage does not require vector cancellation to rescue a violated growth budget.

A falsification control: unseen excursions



A deterministic curve $r(\delta) = \delta + 2\sin(2\pi\delta)$ passes both endpoint checks but exceeds the allowance δ by 2 inside the interval. This control shows why finite secant checks cannot prove a continuous derivative envelope. It is a mathematical example, not another physical trial.

Revised claim and next decisive evidence

An explicit calibration-boundary residual plus a slope allowance repairs the failed origin-only envelope on this corpus. All measured secant-growth checks pass. The ablations support the boundary correction, while reference exponents 0, 0.5 and 1 are indistinguishable by coverage. These results establish a mechanism-supported retrospective case study, not 90% population coverage or validation of a unique weak-turbulence exponent.

Freeze the 0.5-reference rule unchanged before new data collection. Obtain denser powers to test for interior excursions and independently acquired held-out trials, ideally on another apparatus. At least 29 independent successes without failure are needed for a one-sided 95% binomial lower bound above 0.90; this is a best-case minimum, not a powered study design.

Anchor-Symmetry Error Certificate

Feedback-driven rerun | 21 September 2026

Calibration-only theory candidates, rebuilt spectra, and trial-level uncertainty. This supplement reports a new exploratory audit; it does not overwrite the historical figures.

1. Canonical result: 192 mW law ceiling

Three test trials: 250, 300 and 350 mW. Twelve low-power training trials; four above-break law-calibration trials. The table retains the original LOTO anchor-error heuristic for a comparable allowance ablation.

Allowance	Covered trials	CP lower	S > 1	Mean S/e	Min margin
Fitted rate	2/3	0.135	1.000	0.468	-0.298
Historical 2x	3/3	0.368	1.000	0.153	2.114
Theory slope	3/3	0.368	1.000	0.372	0.438
Theory + boundary	3/3	0.368	1.000	0.298	1.345

2. All 22 splits and all 12 training anchors

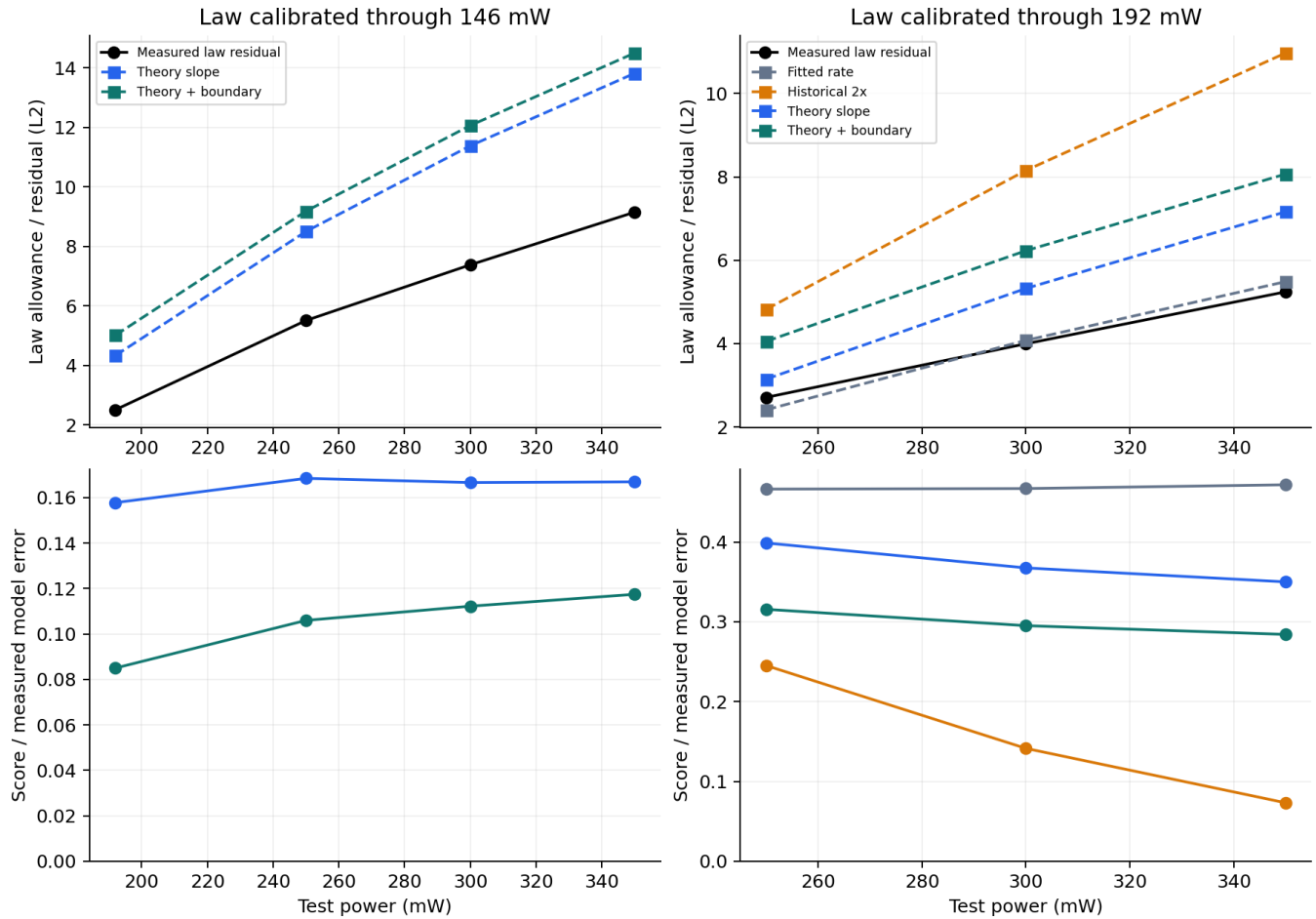
348 configuration rows reuse only three distinct test trials. Pooled fractions are descriptive. The count/CP columns below instead require each trial to pass every swept configuration; they do not form a confidence interval for the pooled rate.

Allowance	Pooled coverage	All-config trials	CP lower	S > 1	Mean S/e
Fitted rate	0.264	0/3	0.000	0.994	0.362
Historical 2x	0.925	1/3	0.017	0.753	0.196
Theory slope	0.296	0/3	0.000	0.991	0.402
Theory + boundary	1.000	3/3	0.368	0.848	0.240

The review's 0.21 baseline is reproduced at the fixed 54 mW anchor: 6/29 configuration rows (0.207; 3 distinct trials, all-config CP lower 0.000). The 0.264 baseline above additionally sweeps anchors.

No row proves 90% population coverage: even 3/3 only gives a nominal lower bound of 0.368. The historical 2x rule was informed by earlier inspection of these test data; it remains a sensitivity comparator.

3. What changes at each test power



Same trained MLP(8), seed 0, and legacy LOTO b throughout. The 146 mW configuration has four distinct test trials; the Setup C fitted rule cannot be estimated there because it requires a three-trial sub-law plus a holdout.

Ceiling	Theory k	Boundary norm	Covered trials	CP lower	Mean S/e
146 mW	15.797	0.682	4/4	0.473	0.105
192 mW	11.926	0.906	3/3	0.368	0.298

The boundary value comes from the already-labelled calibration ceiling and anchor. No test residual enters this value or the theory slope.

4. Separate law adequacy from anchor calibration

The legacy b uses residuals from different leave-one-trial-out models. Its order statistic does not establish split-conformal coverage for the final model at a selected anchor. The labelled-anchor variant instead uses the measured final-model error at that already-labelled anchor. This is exact for the empirical trial-mean target only.

Model / seed	b: LOTO / anchor	Mean S/e: LOTO / anchor	Valid trials: LOTO / anchor
mlp8 / 0	7.003 / 1.487	0.298 / 0.747	3/3 / 3/3
mlp8 / 1	6.970 / 1.475	0.274 / 0.739	3/3 / 3/3
mlp8 / 2	7.144 / 1.588	0.260 / 0.735	3/3 / 3/3
mlp16x16 / 0	5.632 / 1.394	0.000 / 0.000	3/3 / 3/3
mlp16x16 / 1	4.976 / 1.271	0.000 / 0.008	3/3 / 3/3
mlp16x16 / 2	4.752 / 1.458	0.150 / 0.597	3/3 / 3/3
linear / 0	6.366 / 0.721	0.238 / 0.637	3/3 / 3/3

Each validity fraction of 3/3 corresponds to a nominal 95% CP lower bound of 0.368, not seven independent replications of three trials. Law coverage does not depend on model or seed; model sensitivity changes informativeness.

Conditional theory, not a universal guarantee

The capillary spectrum scales as $\text{energy flux}^{(1/2)}$ and $\text{frequency}^{(-17/6)}$. The relevant power slope is 1/2 only under energy flux proportional to drive power. The frequency exponent must not be compared to fitted power exponents.

For $u = \ln(\text{power})$, $r(u) = y_{\text{anchor}} - y(u) - c(u) - c_0$. If $\text{norm}(y'(u) - q_{\text{high}}) \leq k$ over the extrapolation interval, then $\text{norm}(r(u)) \leq \text{norm}(r(u_0)) + k*(u-u_0)$. We test $k = \text{norm}(q_{\text{high}} - 0.5*\text{ones})$.

The derivative assumption is not established by weak-turbulence theory for this apparatus. The experiment audits that candidate; it cannot upgrade it to an unconditional physics or coverage theorem.

5. Paper corrections and related work

Narrow the gap: physics-residual conformal methods can use no target labels. Our distinction is a per-input error lower bound with separate anchor and approximate-law allowances.

Separate training shift from calibration/test exchangeability. OOD relative to training does not automatically invalidate conformal calibration.

Report 348 configurations and three distinct test trials together. A pooled rate over overlapping configurations has no ordinary binomial confidence interval.

Describe the factor of two as retrospective. A round number chosen after viewing test residuals is still test-informed.

Correct reference [7] to Weak turbulence of capillary waves. Correct the first interior 90% finite-sample quantile to $n=19$, not 20. Neither rank arithmetic nor LOTO residuals alone establish exchangeability.

Gopakumar et al. Physics-informed UQ: label-free residual-space calibration. [Source](#)

Wang et al. Symmetry mismatch already yields regression error lower bounds. [Source](#)

Gopakumar et al. Surrogate conformal UQ retains calibration/test exchangeability. [Source](#)

Berman et al. Calibration-error bounds for equivariant functions. [Source](#)

Xu et al. Branched normalizing flows and conditional conformal transport. [Source](#)

Lust and Condurache. GIT transformation-based classification error detection. [Source](#)

Zakharov and Filonenko. Weak turbulence of capillary waves (1967). [Source](#)

Kochurin and Russkikh. Explicit flux and frequency scaling, equations 1-2. [Source](#)

Replacement paragraphs, assumptions and citation distinctions are in PAPER_REVISIONS.md. They do not adopt the reviewer's suggested acceptance-score estimates.

6. Reproducibility and artifact

All rebuilt dataset arrays exactly match the archived arrays: True. Maximum absolute log-spectrum difference: 0.

The source is the supplied 50 MB capillary-wave HDF5, derived from the Orosco, Connacher and Friend UC San Diego deposit (DOI 10.6075/J0WW7HVJ). The repackaging is lossy; this run does not recreate the original 320.7 GB reduction.

Target: average of window-level natural log band-averaged temporal PSDs. Twelve log-spaced bands, 500 Hz-20 kHz; 8192-sample windows, 4096 stride; Welch segments 2048, half overlap, Hann window, linear detrending. All model/law/test partitions are grouped by trial.

Artifact	Contents
RERUN_PROTOCOL.md	Fixed rules and assumptions written before candidate audit
theory_audit.py	Calibration fits, model reruns and row-level audit
rerun/calibration_fits.json	266 fitted configurations, trial lists and frozen allowances
rerun/audit_rows.csv	3176 rows including abstentions, margins and both premises
rerun/results.json	Summary, versions, input/code hashes and rebuild comparison
test_theory_audit.py	Five tests: leakage, boundary, trial counts, abstention and inequality
README_RERUN.md	Commands and interpretation
PAPER_REVISIONS.md	Supported replacement prose and source links

Python 3.8.10; numpy 1.24.4, scipy 1.10.1, scikit-learn 1.3.2, h5py 3.11.0, matplotlib 3.7.5, reportlab 3.6.13

Scope: two canonical ceilings, a 22-split x 12-anchor sensitivity sweep, seven model/seed fits, four fixed allowance rules, two anchor-error choices, and five new verification tests. Historical ensemble benchmarks and earlier figure atlases are not represented as newly rerun results.

Limits: only three or four distinct test trials, one apparatus, already-viewed test data, uncertain drive-to-flux coupling, and empirical trial means rather than a noise-free physical target. Additional independent trials are required to assess the 0.90 population-coverage claim.